

# AI Evaluation & Model Improvement Loop

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Issued 15 Aug 2026 | Due 17 Aug 2026, 6:00 PM IST

Use this HULP workflow as the test bed: an LLM converts messy client conversations into an operator-ready execution brief. The system must preserve corrections, hard constraints and evidence while avoiding invented facts and unnecessary clarification.

**Goal:** build a repeatable, model-agnostic evaluation loop that proves whether a prompt, model or model version is safe and worthwhile to release. The same benchmark should later qualify an open-weight or locally trained model. Required output: intent, brief, facts with source IDs, hard constraints, preferences, missing information, clarification questions, risk flags and confidence.

**Build:** a CLI or small API that runs or cached-replays models, validates JSON, executes graders and produces case-level plus aggregate results.

**Inputs supplied:** Yash\_Pundhir\_Starter\_Cases.jsonl (12), Yash\_Pundhir\_PII\_Context\_Cases.jsonl (4), and Yash\_Pundhir\_prompt\_v0.txt.

## 1 / BENCHMARK AND BASELINE Create the evidence before changing the system.

- Add at least 8 useful cases and lock at least 8 holdout cases before tuning. Run supplied prompt\_v0 unchanged on one primary model; retain raw/cached outputs, settings, latency, tokens and estimated cost.
- Use deterministic/schema graders where possible plus one narrow model grader. Hand-score at least 10 varied outputs, examine disagreements and never rely only on an LLM judge for critical failures.

## 2 / PROMPT-IMPROVEMENT PROOF Show causality, not a better-looking demo.

- For prompt\_v1 and prompt\_v2, keep the primary model/version, schema, settings and locked cases unchanged. Before each edit, record the observed failure and improvement hypothesis.
- Report per-case movements, aggregate improvement, new regressions and remaining critical failures. Define the release threshold you would use and whether v2 actually passes it.

## 3 / PII-AWARE CONTEXT OPTIMISATION Treat context selection as part of model accuracy.

- Names, phones, exact addresses, health/document identifiers, access codes, payment references and saved preferences. Decide which are needed for reasoning, execution-only or prohibited from the model payload. Implement build\_model\_context(input) -> {model\_payload, field\_manifest}; label every field keep / generalise / tokenise / exclude, its purpose and any deterministic downstream attachment.
- Test that excluded values do not reach model requests, caches or logs. On at least four PII cases, compare full versus minimised context for task accuracy, critical failures and token/cost change.

## 4 / MODEL QUALIFICATION AND LOCAL-MODEL PATH Prove that a model change can be evaluated quickly and fairly.

- Hide provider differences behind a model adapter. Freeze prompt\_v2 and the output contract; run the same holdout on at least two materially different model families with comparable settings. Keep model-specific prompts in a separate optimised track.
- Compare accuracy, critical failures, schema reliability, latency and cost. Record the time, code changes and provider-specific branches required to add and qualify the second model; recommend a release decision.
- Submit LOCAL\_MODEL\_PLAN.md: train/dev/final-holdout separation, which reviewed non-holdout failures could become SFT/LoRA or distillation data, leakage controls, resource assumptions and acceptance gates versus the best hosted model. Running an open-weight/local model is a valuable stretch.

## WHAT TO MEASURE

Constraint recall/precision; fact and date accuracy; unsupported-claim rate; superseded-fact errors; clarification precision; schema validity; critical-failure rate.

## SUBMISSION

Repository with prompts, benchmark, model adapter, context builder, graders, tests and cached outputs; REPORT.md with v0->v2 evidence, qualification matrix, PII decisions, failures, learning/AI-tool use and actual time; LOCAL\_MODEL\_PLAN.md; and a 6-minute walkthrough.

## SCORING

**Benchmark/graders 20 | Prompt-improvement proof 20 | Model qualification/local readiness 20 | PII/context optimisation 15 | Engineering/reproducibility 10 | Learning, AI use and ownership 15**

*A partial implementation with sound evidence can score highly. A polished demo without locked comparisons, an inspectable context boundary or a defensible model-release decision cannot.*